Missing Data in RCTs

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Introduction

Missing data are ubiquitous in RCTs and can invalidate inference if not handled carefully. Three missingness mechanisms characterise the problem: **MCAR** (missingness is random), **MAR** (missingness depends on observed variables), and **MNAR** (missingness depends on the unobserved value). Valid analyses require assumptions on the mechanism.

Prerequisites

Basic probability; estimands framework (ICH E9 R1).

Theory

- **MCAR**: $P(\text{missing}) = P(\text{missing} \mid X, Y)$. Complete-case analysis is unbiased but inefficient.
- **MAR**: $P(\text{missing} \mid X, Y) = P(\text{missing} \mid X)$. Multiple imputation, ML, or weighting is unbiased.
- **MNAR**: $P(\text{missing})$ depends on unobserved Y. Sensitivity analyses with assumed MNAR mechanisms are needed.

Missingness is rarely MCAR in practice; MAR is the default operating assumption, with MNAR as sensitivity.

Assumptions

Missingness pattern is characterised by the analyst; auxiliary variables are included in the imputation model; the mechanism assumption matches the method.

R Implementation

```
library(mice); library(naniar)

set.seed(2026)
n <- 300
baseline <- rnorm(n, 5, 1)
arm <- rep(c("ctrl", "trt"), each = n/2)
outcome <- baseline + ifelse(arm == "trt", 1, 0) + rnorm(n, 0, 1)

# MAR: missingness depends on baseline
prob_missing <- plogis(-2 + 0.3 * baseline)
outcome[rbinom(n, 1, prob_missing) == 1] <- NA

df <- data.frame(arm = factor(arm), baseline, outcome)

# Missing-data summary
miss_var_summary(df)

# Multiple imputation
```

```r
imp <- mice(df, m = 10, method = "pmm", printFlag = FALSE)
pool(with(imp, lm(outcome ~ arm + baseline))) %>% summary()
```

Output & Results

Missing-data summary (outcome has ~20 % missingness); pooled MI estimate for treatment effect after adjusting for baseline.

Interpretation

"Under MAR, multiple imputation with 10 imputations gave a treatment effect of 0.94 (95 % CI 0.62-1.26, $p < 0.001$); complete-case analysis gave a similar estimate, consistent with MAR assumption."

Practical Tips

- Prevent missing data by design (pre-specified follow-up, low attrition) before analysis tricks.
- Distinguish missing at random from missing-completely-at-random; CCA needs the stronger MCAR.
- Always report the missingness rate and pattern by arm; differential missingness is a red flag.
- Pre-specify primary analysis under MAR; sensitivity analyses under MNAR (tipping-point).
- ICH E9 R1 estimands framework formalises how missing data interacts with the estimand; align analysis to the estimand, not vice versa.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale